Artifact 21 — Product Requirements Document (PRD)

Execution · Stage A · The Definition — Intent

Source of Truth: This document is the Source of Truth for all Execution Plugin artifacts. A change to any section requires restarting from Stage A. Per CLAUDE.md Article V: "If PRD changes, restart from Stage A."

DOCUMENT DETAILS

Project	Home-Care-AI	**Stage**	Execution → Stage A (The Definition — Intent)
Date	2026-03-27	**Methodology**	8-section PRD template; SMART OKRs; Priority 1/2/3 feature classification
Regulatory	Australian Privacy Act 1988 (APP). HIPAA-grade security as design floor. All data processing in AWS ap-southeast-2 (Sydney). No cross-border transmission in v1.		

Handshake Inputs Consumed

ID	Source	What Transferred
HS-STRAT-01	Artifact 15 §6 (L3 interventions) + §4 (Moment of Truth)	Moment of Truth → P1 feature; all L3 interventions → P1 features (§7.2)
HS-STRAT-02	Artifact 16 (Compliance NFRs)	Mitigation Requirements → Constraints §7.4 and NFR table
HS-STRAT-03	Artifact 20 §12 (Token budget + model selection)	Haiku for L1/L2, Sonnet for L3 → §7.3 Technology
HS-STRAT-04	Artifact 14 §6 "What After" OKRs	OKR-1 through OKR-6 → §4 Objectives (exact, no substitution)
HS-STRAT-05	Artifact 19 §1 Geoffrey Moore statement	→ §3 Background problem definition; §5 market segment

Feeds into

- `prioritization-frameworks` (Execution Stage A, Skill 2) — ranks PRD §7 features
- `agentic-logic-spec` (Execution Stage B, Skill 3) — NFRs + compliance constraints
- `user-stories` (Execution Stage B, Skill 4) — features as testable stories

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1. Summary

Home-Care-AI is a trust-weighted vacancy matching platform for independent Australian home care agencies. When a carer calls in sick, it automatically finds the best replacement — based not only on qualifications and proximity, but on who the client knows and trusts — presents a ranked shortlist to the coordinator for one-tap approval, then handles every notification automatically in the right order.

This PRD covers the **v1 product**: the Smart Match Engine, the Soft Preference Profile (SPP) data model, the 3-Tap Coordinator Approval Flow, and the automated E-3-compliant notification pipeline. Together, these four features eliminate the 11-call phone cascade that currently causes 20% of home care visits to be cancelled.

This document is the Source of Truth for all engineering, design, and QA work in Sprint 1 and beyond. Any feature not in this document is out of scope for v1. Any compliance constraint in this document cannot be overridden without PM Lead approval and a corresponding update to Artifact 17 (Remediation Decision Record).

2. Contacts

Name	Role	Responsibility
PM Lead (founder)	Product Manager + Strategy	Owner of this document. Approves all scope changes.
Angela Morrison (CC-001)	Beachhead Customer — Care Coordinator	Primary user representative. Validates match quality and UX at E1.
Tom Bradley (CC-002)	Beachhead Customer — Care Coordinator	Solo-operator validation. Validates Solo-tier usability.
Engineer	Full-Stack Engineer (Sprint 1)	Builds the system against this PRD. Confirms AX-01 (CRIT-04) and AX-02 (AlayaCare API investigation) before Sprint 1.
Designer	UX/Product Designer	Owns the 3-Tap Approval Flow wireframes and all coordinator-facing screens.
Privacy Counsel	Australian Privacy Law Retainer	Signs off on DPIA completion before v1 data collection begins.
Legal Counsel	Employment + Discrimination Law	E-1 opinion on P-2 (gender preference as matching parameter) — required before matching engine is built.

3. Background

What is this about?

Independent Australian home care agencies send carers to visit elderly clients at home. When a carer calls in sick, the coordinator — often a single person — must find a qualified replacement before the visit starts.

Today, this means making up to 11 phone calls over 30 to 60 minutes. One in five visits is still cancelled. When a visit is cancelled, an elderly client waits alone — sometimes dressed and ready in their chair — and their family finds out when nobody shows up.

The coordinator also carries all the knowledge about what each client needs in her head: which carers the client trusts, who needs a calm introduction, who requires ID at the door, who has difficulty with strangers. When that coordinator leaves, all of that knowledge goes with her.

Home-Care-AI solves both problems. It replaces the 11 phone calls with a 3-tap approval on a phone. And it stores the coordinator's client knowledge in a structured profile that any coordinator can read on Day 1.

Why now?

Three conditions are true simultaneously for the first time:

1	**Customer confirmation (first-party data).** Angela Morrison (CC-001) described the exact workflow she wants: "If the system could detect the absence, find the best available replacement based on qualifications, proximity, and client preferences, propose it to me for approval, and then notify everyone — the replacement carer, the client, and the family — that would save me forty-five minutes per incident." Tom Bradley (CC-002) is cancelling 1 in 5 visits per week. Both have signalled willingness to pay (XP-2A in progress).
2	**A genuine product gap.** AlayaCare's Vacant Visit Agent matches on availability and qualifications only — it has no trust or preference layer. This is confirmed from competitor analysis (Artifact 3). After AlayaCare suggests a candidate, Angela still makes 7 more calls to find someone the client will actually accept. That gap is the product.
3	**Compliance infrastructure is ready to build.** The four critical compliance blockers (Artifact 16) have been resolved (Artifact 17). The product can be built legally and compliantly — with Australian Privacy Act (APP) sign-off, an AU-hosted SMS gateway, and a data model that contains no free-text health fields.

Why this matters to the business

- A validated beachhead (Angela and Tom as paying reference customers) enables peer referral — the primary growth channel in Australia's small, trust-driven home care coordinator community.
- The Soft Preference Profile (SPP) builds a data moat that deepens with every visit logged. At 18 months, switching costs are structural.
- Compliance-by-design is a 6–12 month barrier to entry for any new entrant. Building it first is a strategic advantage, not just overhead.

4. Objective

What are we trying to achieve?

We want every home care coordinator at an independent Australian agency to be able to fill a vacant visit with the right carer — the person the client knows and trusts — in under 5 minutes, with one approval tap, without making a single phone call.

And we want every client's preferences and history to be stored in the system, so that knowledge never lives only in one coordinator's head.

How does this align with the product vision?

From Artifact 12 §1: "Every person receiving home care should be surrounded by people who know them — not just people who are available."

The v1 product is the first step: make sure no vacant visit is filled with the wrong person, and no client sits in their chair waiting for someone who isn't coming.

SMART OKRs (Source: Artifact 14 §6 — HS-STRAT-04, no substitution permitted)

These are the exact "What After" outcomes from the value proposition. They are the success criteria for this PRD.

OKR	Outcome	Target	Baseline	Measured By	Time-Bound
OKR-1	Vacancy time-to-fill	**< 5 minutes** per incident	30–60 min (Angela CC-001)	E1 concierge stopwatch (XP-1A)	Within 30-day trial
OKR-2	Cancellation rate	**< 2%** per agency per week	~20% (Tom CC-002)	Incident log	Within 30-day trial
OKR-3	Coordinator 1-tap trust	≥ **70%** of top-ranked candidates approved without verification calls	0% (no system exists)	Deviation log (XP-1B)	Steady-state (post-SPP migration)
OKR-4	SPP completeness	≥ **80%** of active clients with ≥ 3 SPP fields populated	~0% (sticky notes only)	SPP completeness score (P-10)	Within 90 days of onboarding
OKR-5	Family notification rate	**100%** of confirmed replacements trigger automated family notification	Ad hoc / ~0% (Arthur Kovacs failure case)	Audit log ACT-F-01 delivery confirmation	v1 launch
OKR-6 ■■	Knowledge survivability	New coordinator effective in **< 5 days** with SPP handover	2–3 months shadowing	Agency owner assessment at 30 days	Within 30 days of coordinator onboarding

■■ **OKR-6 note:** Baseline is directional — not sourced from a named agency owner interview. Will be updated after XP-2A agency owner interviews. OKR-6 does not gate Sprint 1.

How will success benefit customers and the business?

Coordinators (CC): Angela goes from a 45-minute panic every morning to a 5-minute task. She stops making 11 calls she didn't want to make. She stops forgetting to call the family. She stops worrying that the next coordinator won't know what she knows.

Agency owners: Client preferences become a product asset — not a people dependency. The agency survives a coordinator departure. The agency owner stops taking furious calls from families who found out when nobody showed up.

Business: Two validated reference customers become peer referrals. Peer referral is the primary growth channel in a small professional community. The SPP moat deepens with every active agency. Gross margin crosses 60% at ~83 agencies (Artifact 20 §07.2).

5. Market Segments

Who are we building this for?

Primary — Care Coordinator (CC) ■ Beachhead

Home care coordinators at independent Australian agencies with 20–200 active clients. One to three coordinators per agency. Sole operator (Tom: 25 clients, 1 coordinator) through team lead (Angela: 60+ clients, 3 coordinators).

They work under constant schedule pressure. They operate on their phone. They carry all client preference knowledge in their heads. They are personally responsible for both care quality and compliance — with no system that unifies both.

Entry point: The coordinator is the product champion. They feel the pain directly, every morning. The agency owner is the budget approver — they enter later (XP-2A).

Defining constraint: The product must work on a 5-inch phone screen, one-handed, at 6:30 AM, before coffee. Every design decision is evaluated against this constraint (Artifact 15 DR-5). Any feature that requires a desktop, a login process longer than 15 seconds, or more than 3 taps to complete the core action is out of scope for the coordinator's primary workflow.

Secondary — Agency Owner / Operations Manager

Owners and operations managers who depend on 1–3 coordinators to maintain care quality for 20–200 clients. Often the founder. Personally liable for care quality and compliance under the Aged Care Quality Standards.

Entry point: Agency owners become active buyers after seeing the coordinator use the product during E1 (concierge phase). They are not the champion — the coordinator is — but they sign the purchase order.

v1 value: Institutional knowledge is no longer a person dependency. The SPP is the agency's knowledge record, not the coordinator's memory.

Future — Family Contact (FC) [NOT IN v1]

Unlocks after CC adoption is validated and the E-3 notification gate is live. The family contact receives an automated SMS after a replacement is confirmed — before they have to call the agency. Not in scope for v1.

Future — Home Care Nurse / Case Manager (HCN) [NOT IN v1]

Unlocks after SPP and continuity history data is live. Not in scope for v1.

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6. Value Proposition

What problem do we solve — and better than the alternatives?

(Source: Artifact 14 §1, Artifact 19 §1. HS-STRAT-05 transfer from Artifact 19 positioning.)

For home care coordinators at independent Australian agencies managing 20–200 clients, who lose 30–60 minutes and cancel 1 in 5 visits every time a carer calls in sick — because no system knows which replacement the client will actually open the door for — Home-Care-AI is a trust-weighted vacancy matching platform that fills every vacant visit in under 5 minutes with the right carer for that client, and automatically notifies the carer, the client, and the family in the right order — so that no senior sits in their chair waiting, and no family finds out before their parent does.

Unlike AlayaCare's Vacant Visit Agent, which matches on availability and qualifications only, Home-Care-AI matches on who the client trusts — using a Soft Preference Profile built from the coordinator's institutional knowledge — so the coordinator approves once, and the right person arrives.

What do customers gain?

Gain	Evidence
45 minutes back per incident	Angela (CC-001): "Three incidents a week — that's over two hours of panicked phone calls I don't have to make."
Zero cancelled visits (target < 2%)	From Tom's ~20% cancellation baseline — trust-matching surfaces candidates manual calls miss.
No more Arthur Kovacs failure cases	E-3 gate ensures client is notified before family, automatically, every time — not when Angela remembers.
Client knowledge that survives staff turnover	Angela: "If I get hit by a bus tomorrow, half the knowledge about our clients walks out the door with me." SPP eliminates that single point of failure.
Every decision logged and auditable	Every replacement decision written to an immutable audit log. No more personal liability for undocumented choices.

What pains do we take away?

Pain	Current State	After v1
11-call morning cascade	30–60 min, 3–5 times per week	3 taps, < 5 minutes
Mental triage (who trusts who)	Angela's memory — fragile, untransferable, unreliable at 6:30 AM	SPP triage panel: instant, accurate, available to any coordinator
Forgotten client/family notification	Ad hoc — forgotten in the cascade	Automated, E-3 compliant, structured in code
Knowledge locked in one person's head	Sticky notes and memory walk out the door with the coordinator	SPP: structured, searchable, transferable from Day 1
No audit trail	Undocumented, reconstructed from memory if challenged	Immutable, append-only, 7-year retention

How are we better than the alternatives?

Factor	Manual Calls	AlayaCare Vacant Visit	Home-Care-AI v1
Time to fill	30–60 min	Faster — but trust-blind	**< 5 min**
Trust / preference matching	High (coordinator knows) — but unsystem atised	None	**High — SPP-driven, transferable**
Family notification	Forgotten	None	**100% automated, E-3 gated**
Institutional knowledge capture	Sticky notes + memory	Partial — custom fields, no matching	**Full SPP — structured, matched, transferable**
APP compliance	Implicit	Partial	**DPIA-complete, by design**
Works standalone (no EMR needed)	Yes	No (preferred)	**Yes — standalone v1**

7. Solution

7.1 UX / Prototypes

Primary workflow — Exception Handling (vacancy incident)

The coordinator's primary workflow is a 10-step sequence from absence detection to incident resolution (Artifact 15 §3). The product's job is to collapse Steps 1–9 from 30–45 minutes to under 10 minutes, while keeping the coordinator in control of the single most important decision: who goes to which client.

Screen sequence (v1 coordinator app, mobile-first)

```
1. [PUSH NOTIFICATION]
   "Jenny called in sick — 3 visits affected.
   Replacements ready for review."
   [Tap to open → ]

2. [VACANCY OVERVIEW SCREEN]
   Three visit cards: Mrs. Kim (8:00 AM) · Arthur Kovacs (9:30 AM) · Lin Chen (11:00 AM)
   Each card shows: Client name · Visit time · SPP summary (familiarity threshold label)
   [Tap Mrs. Kim's card → ]

3. [CANDIDATE SHORTLIST — Mrs. Kim]
   TOP MATCH: David Kim
   ✓ Qualified · 2.1 km · 2 prior visits with Mrs. Kim
   [Secondary: Sarah Ng — ✓ Qualified · 4.3 km · 0 prior visits]
   [Third: Marco Torres — ✓ Qualified · 5.8 km · 1 prior visit]
   [Tap David Kim → ]

4. [★ MOMENT OF TRUTH — COORDINATOR APPROVAL CARD]
   David Kim
   ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
   ■ ✓ Qualified for Mrs. Kim's care ■
   ■ ■ 2.1 km — Surry Hills ■
   ■ ★ 2 prior visits with Mrs. Kim ■ ← FAMILIARITY FLAG (largest element, DR-1)
   ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
   If you approve, this is what will be sent:
   → David: "CareBridge needs you to cover a visit in Surry Hills at 8 AM. Reply YES/NO."
   → Mrs. Kim: "Your visit today will be with David Kim, who has visited you before."
   → Margaret (daughter): "Your mother's visit today will be covered by David Kim..."
      [Sent AFTER Mrs. Kim is notified — guaranteed]
   [APPROVE] ← single tap

5. [RESOLUTION SCREEN]
   ✓ David Kim assigned — awaiting reply
   ✓ Mrs. Kim notified — SMS delivered
   ✓ Margaret Chen notified — SMS delivered
   "This incident has been logged with timestamp and decision rationale."
   [Done]
```

Design requirements (binding — from Artifact 15 §7)

Ref	Requirement	Why Non-Negotiable
DR-1	Familiarity flag ("2 prior visits with Mrs. Kim") is the largest element on the candidate card — above the match score	This single element is the trust trigger. A score ("94% match") without context creates verification calls; a human fact creates approval. Per Moment of Truth (Artifact 15 §4).
DR-2	Notification preview (what will be sent to David, Mrs. Kim, and Margaret) appears before the Approve tap	Angela must see downstream consequences before committing. Removes first-use anxiety. Makes E-3 constraint visible as a promise, not a background rule.
DR-3	Resolution screen is emotionally satisfying — client-centred language ("Mrs. Kim has been notified and knows David is coming"), not system language ("ACT-P-01 delivered")	This is the product's emotional payoff — the closure Angela never gets from 11 phone calls. Design investment here is proportionate to the product's core promise.

Ref	Requirement	Why Non-Negotiable
DR-4	SPP empty state feels like an invitation ("Add Mrs. Chen's preferences now — 2 min"), not a dead end	Empty SPP = empty briefing = product feels worse than calling. Empty state must convert coordinator anxiety into a population action.
DR-5	Full 3-Tap Flow (candidate review → approval preview → confirm) completable one-handed on a 5-inch phone screen in under 60 seconds	Angela is standing in her kitchen at 6:30 AM. No desktop. No form entries. No text fields. If it requires a 4th tap for any reason, justify it to PM Lead before merging.

7.2 Key Features

Feature priority classification:

P1 — **P1 — Must ship v1:** Product does not exist without these. A Level 3 intervention absent from this list breaks the safety chain (CLAUDE.md HS-STRAT-01). An OKR without its P1 feature cannot be measured.

P1.1 — **P1.1 — Post-beachhead (60 days after first paying customer):** Specifically requested by beachhead users; deferred to protect Sprint 1 focus.

P2 — **P2 — v2 (after beachhead validation + compliance unlock):** Requires a v2 unlock condition per Artifact 17 §07.

[P1] Feature 1: Soft Preference Profile (SPP) Data Model + Capture Flow

What it is: A structured, per-client knowledge record. Replaces sticky notes and coordinator memory with a database that any coordinator can read on Day 1.

v1 schema (structured fields only — no free text anywhere)

Field	Type	Description	Compliance Note
P-1	Client identity (pseudonymised)	UUID + first name only in app	PHI — pseudonymised
P-2	Gender preference	Categorical dropdown (Female / Male / No preference)	Advisory only — NOT in scoring algorithm (CRIT-03). Displayed as "Client preference (advisory — not scored)."
P-3	Familiarity threshold	Categorical: Known carers only / Briefed-acceptable / Any	Key matching parameter. Drives client notification phrasing (G-DS-05).
P-4	Cultural/religious considerations	Structured multi-select (dietary, observance, language)	Yellow zone — explicit opt-in required.
P-5	Personal sensitivities	Structured max-100-char field (e.g. "Do not move items in lounge")	Not free-text health information — structured, coordinator-entered.
P-6	Entry protocol (replacement)	Dropdown: ID check required / Introduction call first / No special requirement	Operational, not clinical.
P-7	Visit history / familiarity	Auto-populated: carer ID, visit count, last visit date	Source of "2 prior visits" familiarity flag.
P-8	Binary carer flags	Per-carer flag: "Client has accepted this carer before"	Used in shortlist scoring.
P-10	SPP completeness score	Calculated: % fields populated / total fields	OKR-4 measurement input.

Field	Type	Description	Compliance Note
P-11	Client suburb	Text (suburb only — not full address)	Used for proximity scoring only. Not disclosed to carer.

What it is NOT: A clinical record. A free-text notes field. A health diagnosis store. P-9 (free-text SPP notes) is explicitly excluded from v1 (CRIT-04 resolution, Artifact 17 §05). No exceptions.

Capture flows

- **ACT-S-01:** SPP entry at client intake — coordinator-guided structured form
- **ACT-S-02:** Concierge-assisted SPP migration session (coordinator + PM Lead for existing clients). **Note (B3-B):** ACT-S-02 uses AlayaCare's existing read-only CSV export — it does NOT require write-back capability. AX-02 outcome does not affect this flow. If AX-02 confirms write-back unavailable, ACT-S-02 is replaced by Artifact 24 US-15 (self-service CSV import) for Sprint 2.
- **ACT-S-03:** Visit-triggered completeness prompt — system surfaces empty fields after a vacancy incident involving that client ("Add Lin Chen's preferences now — it will help next time")

OKR it measures: OKR-4 (≥ 80% of active clients with ≥ 3 SPP fields populated within 90 days).

[P1] Feature 2: Smart Match Engine

What it is: An algorithm that, when a vacancy is recorded, automatically ranks available and qualified carers in order of suitability for the specific client — in under 30 seconds.

Algorithm (rule-based in v1 — no ML, fully auditable)

```
STEP 1 — Qualification Gate (hard binary — pass/fail):
  → Carer credentials match client care requirements
  → No expired certifications
  → Available (not already scheduled at this time)
  → Only carers who pass this gate appear in the shortlist

STEP 2 — Proximity Score (weighted):
  → Google Maps Distance Matrix API: carer suburb → client suburb
  → Lower distance = higher score
  → Suburb only — no GPS tracking, no full address disclosure (APP compliant)

STEP 3 — SPP Match Score (weighted — the trust layer):
  → P-7 familiarity history: prior visit count with this specific client (highest weight)
  → P-8 binary acceptance flag: client has accepted this carer before
  → P-3 familiarity threshold: "known only" clients de-rank unfamiliar carers
  → P-2 (gender preference): NOT in scoring function (CRIT-03 compliance guard)

FINAL RANK: Qualification pass + highest (Proximity + SPP Match) = #1 candidate
```

Algorithm constraint (binding from Artifact 17 §04)

```
E1_LEGAL_SIGNOFF: bool = false // default until legal sign-off obtained
IF E1_LEGAL_SIGNOFF == false:
    ASSERT P2 NOT IN scoring_weights
    DISPLAY P2 to coordinator as advisory: "Client preference (advisory — not scored)"
    LOG guard_id = 'G-CC-4', guard_passed = true
```

Actions in this feature

- ACT-V-01: Detect vacancy event (absence recorded by coordinator or triggered by carer)
- ACT-V-02: Run qualification gate against carer database
- ACT-V-03: Compute proximity score (Google Maps API — APP 8 review required before production — SC-07 action item)
- ACT-V-04: Compute SPP match score
- ACT-V-05: Generate and send coordinator push notification ("3 replacements ready")

- ACT-V-06: Serve ranked shortlist to coordinator app
- ACT-V-07: Display match explanation (SPP field labels that contributed — no full SPP disclosure, CC-6 guard)

OKR it measures: OKR-1 (< 5 min time-to-fill), OKR-2 (< 2% cancellation rate), OKR-3 (≥ 70% 1-tap trust).

[P1] Feature 3: 3-Tap Coordinator Approval Flow — The Moment of Truth

What it is: The single most important feature in the product. Angela reviews the shortlist, taps her chosen candidate, sees the full notification preview, and taps Approve. The system executes everything after that tap — without Angela touching the phone again.

This is the HITL gate. It cannot be bypassed. It cannot be automated away. Every L3 action downstream of this tap requires this tap first.

Actions

- ACT-A-01: Serve the coordinator approval card (familiarity flag + qualifications badge + notification preview)
- ACT-A-02: Record coordinator approval (sets `coordinator_approved = true`, writes HITL_CONFIRMED audit entry)

What ACT-A-01 must display (non-negotiable UI elements — from HS-STRAT-05h): 1. Familiarity count: "2 prior visits with Mrs. Kim" — this exact format, this exact prominence 2. Qualifications confirmed badge: "✓ Qualified for [client first name]'s care requirements" 3. Full notification preview: what message David will receive, what Mrs. Kim will receive, what the family will receive — in that order, with the E-3 note: "[Family name] will be notified after [client name]"

What the approval card must NOT contain

- Raw match percentage or score as the primary element (destroys trust — Angela sees a number with no context)
- Any SPP field that discloses sensitive health information to the coordinator in this context (CC-6 guard: the briefing assembly, not the approval card, is where SPP details flow to the carer)
- The reason P-2 was or was not applied (G-CC-4 guard: P-2 advisory is shown separately, not embedded in match rationale)

OKR it measures: OKR-3 (≥ 70% 1-tap trust rate, measured by XP-1B deviation log).

[P1] Feature 4: Automated Carer Assignment + Briefing

What it is: After coordinator approval, the system sends the replacement carer two messages: (1) the assignment request and (2) the client briefing. Both are template-based (no LLM). Both require prior coordinator approval (ACT-A-02 = true).

ACT-C-01 — Carer Assignment SMS: Template: `"Hi [carer_first_name], [agency_name] needs you to cover a visit for a client in [client_suburb] at [visit_time]. Reply YES to confirm or NO if unavailable."`

What ACT-C-01 must NOT include (CC-8 guard)

- Client full name
- Client full address
- Any SPP data
- Any health information

ACT-C-02 — Carer Briefing SMS (sent after carer confirms): Template assembled from SPP structured fields:

- Entry protocol label (P-6): e.g. "Please knock and wait — client takes a moment to answer"
- Personal sensitivities (P-5): e.g. "Please don't move any items in the lounge"
- Familiarity phrasing (P-3): If threshold = "known carers only", add "This client prefers familiar carers — please introduce yourself clearly and take a calm, steady approach."

ACT-C-02 guard (CC-6, mandatory)

```
ASSERT match_explanation_in_payload = false
ASSERT gender_preference_in_payload = false
IF SPP empty for this client:
    SEND minimal briefing ("Client prefers a calm, steady approach")
    ALERT coordinator: "Mrs. Chen's preferences haven't been added yet. Add them now (2 min)."
```

Failure handling

- ACT-C-01 delivery failure → L1 alert to Angela: "SMS to David failed — call him directly by [visit_time - 30 min]"
- Carer declines (replies NO) → re-surface shortlist with next candidate, do not auto-assign
- Carer no-reply after 15 minutes → L1 alert to Angela: "David hasn't replied — confirm or pick another candidate"

[P1] Feature 5: Client + Family Notification Pipeline with E-3 Gate

What it is: After coordinator approval (and carer has confirmed), the system automatically sends an SMS to the client and then to the family — in that exact order, guaranteed in code.

> **The E-3 gate is a structural code constraint, not a policy. It cannot be bypassed by any configuration, user action, or edge case. This is the Arthur Kovacs constraint.**

ACT-P-01 — Client Notification SMS: Template (phrasing branch on P-3 familiarity threshold):

- If P-7 visit count > 0 (carer has visited before): `"Good morning, this is [agency_name]. Your visit today will be with [carer_first_name], who has visited you before. They'll arrive at [visit_time]."`
- If P-3 = "briefed-acceptable" AND visit count = 0: `"Good morning, this is [agency_name]. Your visit today will be covered by [carer_first_name]. They know about your preferences and will introduce themselves when they arrive at [visit_time]."`
- If P-3 = "known carers only" AND carer is unfamiliar: DO NOT assign this carer. This client should not reach ACT-P-01 with an unfamiliar carer — the shortlist algorithm should have excluded them. If this state is somehow reached → escalate to coordinator as VACANCY_UNRESOLVED.

ACT-P-01 note on P-3 ≥ 2 (cognitively impaired clients): G-DS-05 guard: clients with P-3 familiarity threshold = "known carers only" must never receive the phrase "who has visited you before" if the assigned carer is not actually known. The phrasing branch above enforces this, but the `agentic-logic-spec` must assert this guard explicitly.

E-3 Gate 1

```
BEFORE ACT-P-01:
    ASSERT coordinator_approved = true
    SET client_notified = false

AFTER ACT-P-01 delivery confirmed:
    SET client_notified = true
```

```
IF ACT-P-01 delivery fails:
    SET client_notified = false
    ALERT coordinator: "[client_first_name] has no notification channel — please call them directly before family is notified."
```

ACT-F-01 — Family Notification SMS: Template: `"Hi [family_contact_first_name], [agency_name] is letting you know that [client_first_name]'s visit today will be covered by [carer_first_name], who has visited [client_pronoun] before. Arranged by [coordinator_first_name]. — [agency_name]"`

E-3 Gate 2 (the Arthur Kovacs constraint — non-bypassable)

```
BEFORE ACT-F-01:
    ASSERT client_notified = true
    IF client_notified = false:
        SUPPRESS ACT-F-01
        ALERT coordinator: "Family notification held — [client_first_name] has not yet been notified. Notify them manually to release."
        LOG event_type = 'E3_GATE_BLOCKED', guard_id = 'G-E3-1'
```

OKR it measures: OKR-5 (100% of confirmed replacements trigger automated family notification).

[P1] Feature 6: Immutable Audit Log

What it is: Every state transition in every vacancy incident is written to an immutable, append-only audit log. This is not a feature the coordinator notices — it is the product's compliance and trust infrastructure.

Why non-negotiable: Per CLAUDE.md Article IX: "If a state transition happened and was not logged, it did not happen (for compliance purposes)." SD-01B (audit log completeness) is the merge gate — no feature ships if its state transitions are not 100% logged.

Audit log schema: Full `HIPAAAuditLogEntry` schema from CLAUDE.md Article IX, extended with APP-specific fields (Artifact 16 §7):

- `cross_border_disclosure`: false (all data stays in ap-southeast-2)
- `app8_basis`: "Domestic — no cross-border transmission"
- `consent_record_id`: link to consent record where applicable
- `guard_id`: which compliance guard fired (G-CC-4, G-DS-05, G-E3-1, CC-6, CC-8)
- `guard_passed`: boolean

Minimum state transitions that must be logged per vacancy incident

Event	State Before	State After
Vacancy detected	NORMAL	ALERT_PENDING
Coordinator notified (shortlist sent)	ALERT_PENDING	HITL_PENDING
Coordinator approved	HITL_PENDING	ALERT_DISPATCHED
Carer SMS sent (ACT-C-01)	ALERT_DISPATCHED	ALERT_DISPATCHED
Carer confirmed	ALERT_DISPATCHED	ALERT_DISPATCHED
Client notified (ACT-P-01)	ALERT_DISPATCHED	ALERT_DISPATCHED
Family notified (ACT-F-01)	ALERT_DISPATCHED	ALERT_DISPATCHED
Incident resolved	ALERT_DISPATCHED	RESOLVED
Vacancy unresolved	any	VACANCY_UNRESOLVED
E-3 gate blocked	ALERT_DISPATCHED	ALERT_DISPATCHED (with guard_id E3_GATE_BLOCKED)

Infrastructure: AWS CloudTrail + write-once S3 (ap-southeast-2). 7-year retention. No application role may DELETE or UPDATE a log entry. Any role with UPDATE access on the log table is a CRITICAL security finding.

[P1] Feature 7: Vacancy Unresolved Escalation

What it is: When no suitable candidate is found (all shortlist candidates decline or are unavailable), the system escalates to the coordinator as VACANCY_UNRESOLVED — it does not auto-assign, it does not pick a fourth option silently, and it does not leave the incident in a hanging state.

VACANCY_UNRESOLVED Protocol

```
IF all shortlist candidates decline OR timeout:
    SET incident_state = VACANCY_UNRESOLVED
    ALERT coordinator (L3 escalation — Sonnet model, see §7.3):
        "No suitable replacement found for [client] at [visit_time].
        Candidates tried: [list]. Reason each declined: [if provided].
        Options: Extend search criteria / Call manually / Mark as cancelled."
    LOG event_type = 'VACANCY_UNRESOLVED', state_after = 'VACANCY_UNRESOLVED'
```

The system NEVER auto-cancels a visit without coordinator confirmation. VACANCY_UNRESOLVED is a signal to the coordinator — the decision is theirs.

Why this is P1: Per CLAUDE.md Article VIII: "The system must never be left in a hanging ALERT_PENDING state." VACANCY_UNRESOLVED is the resolution path for the failure case that currently produces the 20% cancellation rate.

[P1.1] Feature 8: Compliance Gap Dashboard

What it is: Angela's exact specification (Artifact 2c): *"Here are the seventeen things that are out of compliance. Here are the five that are critical."*

A real-time prioritised list of compliance items: overdue care plan reviews, expiring carer credentials, visit documentation gaps. One-tap remediation path.

Why deferred: Sprint 1 focus must be on the replacement-matching flow (Features 1–7). The compliance dashboard adds significant data model work and is not required to validate OKR-1 through OKR-5. Deferred to post-beachhead to avoid Sprint 1 scope inflation.

Trigger for activation: OKR-1 and OKR-3 both validated in E1 (Angela confirms time-to-fill < 5 min AND ≥ 70% 1-tap approval rate).

[P2] Feature 9: WhatsApp Notifications

Unlock condition (Artifact 17 §07): APP 8 legal review completed — Privacy Counsel confirms provider's DPA satisfies APP 8 "substantially similar" protection. OR Carer App in-app push is ready (preferred v2 path — replaces SMS entirely, no APP 8 issue, and reduces COGS by 99% per Artifact 20 §13).

Why not v1: WhatsApp routes via Meta's US-hosted servers — APP 8 cross-border disclosure without confirmed legal basis. Resolved by AU-hosted SMS gateway for v1 (Artifact 17 CRIT-01).

[P2] Feature 10: AlayaCare Bi-Directional Integration

Unlock condition: Engineer confirms AlayaCare Connect write-back endpoint availability (AX-02, due 2026-04-01). If read-only only, integration is deferred. If bi-directional confirmed, integration scoping begins post-beachhead.

7.3 Technology

Infrastructure stack: AWS ap-southeast-2 (Sydney). No data leaves Australia. Chosen for APP 8 compliance (no cross-border trigger), not cost.

Component	Service	Why
State machine + agentic orchestration	AWS Lambda (event-driven)	Serverless; scales to zero at beachhead; supports step function for vacancy incident workflow
SPP + carer data (structured)	Amazon DynamoDB	Low-latency reads for shortlist generation (< 200ms confirmed — F-4, Artifact 7); HIPAA-grade encryption at rest
Audit log (immutable)	AWS CloudTrail + write-once S3	Append-only by infrastructure design; 7-year retention; no application can UPDATE or DELETE
Proximity scoring	Google Maps Distance Matrix API	Proven accuracy for Australian geography (F-3 confirmed). APP 8 review required before production (AX-01 action item from SC-07 — Artifact 10 §9). Sends carer postcode only — no client address.
SMS notifications (AU-hosted)	MessageMedia OR AWS SNS (ap-southeast-2)	Domestic processing — no APP 8 trigger. Evaluate AWS SNS pricing vs. MessageMedia at 20+ agencies (Artifact 20 §10 Recommendation 2). Channel Wrapper architecture: SMS is a configuration parameter, not hardcoded.
Encryption + key management	AWS KMS (ap-southeast-2)	SSE-KMS for all stored data. 1 customer-managed key per agency.

AI model selection (binding from HS-STRAT-03, Artifact 20 §12)

Action Type	Model	Token Budget	Cost/Action	Constraint
L1/L2 routine orchestration (all state management, HITL coordination)	**Claude Haiku 4.5** (`claude-haiku-4-5-20251001`)	2,200 tokens	~$0.0015	Prompt caching enabled Day 1. Cache static system prompt blocks.
L3 — VACANCY_UNRESOLVED escalation	**Claude Sonnet 4.6** (`claude-sonnet-4-6`)	3,500 tokens	~$0.0165	Sonnet reserved exclusively for this action. Any new L3 classification requires PM Lead sign-off.
Notification content (ACT-C-01, ACT-C-02, ACT-P-01, ACT-F-01)	**No LLM — template interpolation**	0 tokens	$0	CRIT-02 compliance constraint. Cannot be changed without Artifact 17 amendment + Privacy Counsel sign-off.
Audit log entries	**No LLM**	0 tokens	$0	Structured write. APPAuditLogEntry schema is deterministic.

All v1 external-facing content (notifications, briefings) uses template-based string interpolation. No LLM generates text transmitted to carers, clients, or families in v1. Any code that passes notification text through an LLM API is a CRITICAL compliance finding.

7.4 Assumptions + Constraints

Assumptions (what we believe but haven't fully proven)

ID	Assumption	Status	Test
V-1	Coordinators trust a machine-ranked shortlist enough to approve without verification calls	■ **Pending**	XP-1A/1B — E1 live data

ID	Assumption	Status	Test
V-2	The familiarity flag ("2 prior visits") is the trust trigger — not the match score	■ **Pending**	E1 first-use observation (deviation log)
V-3	SPP can be populated at sufficient completeness (≥ 80%) to differentiate match quality	■ **Pending**	XP-3A — self-serve template
VI-1	Agency owners will pay for a standalone product without EMR integration	■ **Pending**	XP-2A — LOI meetings
F-1	Carers reply to SMS assignment requests within 15 minutes at ≥ 70% rate	■ **Pending**	XP-4A — SMS dry run
S-3	SPP data is sticky enough that switching cost > 4 weeks of rebuild	■ **Pending**	XP-3B — moat probe
E-1	P-2 (gender preference) is lawful as a matching parameter	■ **Legal review pending**	Employment + discrimination law counsel opinion
AX-01	P-9 (free-text field) is confirmed absent from v1 data schema	■ **Engineering confirmation pending**	Engineer confirms in writing before Sprint 1
AX-02	AlayaCare Connect supports bi-directional write-back	■ **Engineering investigation pending**	API documentation review (due 2026-04-01). **Branching plan (B3-C):** If write-back confirmed → AlayaCare integration scoping enters v2 roadmap. If write-back unavailable → Artifact 24 US-14 (Manual Absence Record) and US-15 (Bulk SPP Import via CSV) enter Sprint 2 backlog; v1 operates fully standalone. Either answer is fine — the unknown is the risk.

Compliance Constraints (binding from Artifact 16 + Artifact 17 — cannot be overridden)

CRITICAL — must be implemented before Sprint 1 data collection begins

ID	Constraint	Source	Implementation Rule
CRIT-01	All v1 notifications via AU-hosted SMS gateway only. No WhatsApp/Twilio/Meta-hosted channel in v1.	Artifact 17 §02	Channel Wrapper architecture. `channel_config.v1 = 'SMS_AU'`. Hardcoding any channel is prohibited.
CRIT-02	All external content (notifications, briefings) generated by template interpolation only. No LLM call generates text sent to carers, clients, or families.	Artifact 17 §03	`content_generation_method == TEMPLATE_INTERPOLATION` asserted in agentic-logic-spec. Any LLM assignment to notification actions = CRITICAL defect.
CRIT-03	P-2 (gender preference) carries zero weight in the matching algorithm. Displayed as advisory only: "Client preference (advisory — not scored)."	Artifact 17 §04	`E1_LEGAL_SIGNOFF = false` constant in code. `ASSERT P2 NOT IN scoring_weights`. Guard logged as G-CC-4.
CRIT-04 (PENDING AX-01)	P-9 (free-text SPP notes) does not exist in the v1 schema. No unstructured text field of any name. Engineer must confirm in writing before Sprint 1.	Artifact 17 §05	AX-01 engineering confirmation before any data model is finalised. Schema review at PR gate.

HIGH — must be resolved before GA (any feature shipping to production)

ID	Constraint	Source	Blocks
HIGH-01	Consent framework for sensitive information (P-2, P-3, P-4, P-5) not yet specified.	Artifact 16 §5	GA
HIGH-02	Lawful basis for carer SPP disclosure (ACT-C-02 briefing) not yet documented.	Artifact 16 §5	GA
HIGH-03	DPIA for CC-1 compound (P-3 + P-4 + P-5 combination) not yet initiated.	Artifact 16 §5	GA
HIGH-04	Google Maps API APP 8 review not confirmed. ACT-V-03 transmits carer postcode to Google's US-hosted servers.	Artifact 16 + Artifact 10 SC-07	Production use of ACT-V-03
HIGH-05	RAG vector store isolation not architecturally specified. (v2 feature — not a Sprint 1 blocker, but must be specified before any RAG build begins.)	Artifact 16 §5	Any RAG build
HIGH-06	Cross-session coordinator memory isolation not specified.	Artifact 16 §5	Sprint 1
HIGH-07	CC-6 guard (match explanation absent from carer briefing) not yet in code.	Artifact 16 §5	Sprint 1
HIGH-08	E-3 structural gate not yet in code.	Artifact 16 §5	Sprint 1

HIGH-06, HIGH-07, and HIGH-08 are Sprint 1 requirements — they block sprint completion, not just GA.

High-Risk Compound Combinations (from Artifact 16 §4 — per CLAUDE.md Article IV Rule 7)

These must appear in the PRD Constraints so engineers cannot unknowingly build illegal data flows:

CC ID	Combination	What It Creates	PRD Constraint
CC-1	P-3 (familiarity threshold) + P-4 (cultural/religious) + P-5 (personal sensitivities) combined in any report, API response, or notification	Near-complete cognitive vulnerability profile — clinical monitoring classification under APP	Do NOT combine CC-1 fields in any single API response, notification payload, or report. If combined, must trigger `CC1_COMPOUND_BLOCKED` event and DPIA gate.
CC-2	ACT-P-01 client notification phrasing + P-3 familiarity threshold	For P-3 = "known carers only" (proxy for cognitive impairment), specific phrasing creates undue distress	G-DS-05 guard: "who has visited you before" phrase BLOCKED for P-3 ≥ 2 clients. Phrasing must branch (Feature 5 above).
CC-3	P-3 + P-5 + P-7 in shortlist display	Full cognitive profile reconstructible	Shortlist displays P-7 visit count as "X prior visits" — not P-3 label or P-5 content.
CC-4	Carer briefing (ACT-C-02) + P-2 + P-3 + P-5	Health information disclosure to carer without lawful basis	CC-6 guard mandatory on all briefing payloads (Feature 4 above).
CC-5	Any combination of carer ID + client ID + visit history in a single plaintext log entry	PHI correlation without pseudonymisation	Audit log uses UUIDs only — no first names, no DOBs, no addresses in any log field.

8. Release Plan

What goes in v1?

v1 is the smallest thing that delivers the coordinator's core job: fill a vacant visit with the right carer, in under 5 minutes, with one tap, and notify everyone automatically in the right order.

In v1

- SPP data model + capture flow (P1)
- Smart Match Engine — rule-based (P1)
- 3-Tap Coordinator Approval Flow (P1)
- Carer SMS assignment + briefing (P1 — L3)
- Client SMS notification with E-3 gate (P1 — L3)
- Family SMS notification with E-3 gate (P1 — L3)
- Vacancy Unresolved escalation (P1)
- Immutable audit log — every state transition (P1)
- Mobile-first coordinator app — iOS/Android (P1)

In v1.1 (60 days post-launch, post-OKR-1/OKR-3 validation)

- Compliance gap dashboard ("seventeen things out of compliance" — Angela's request)
- SPP empty state population flow improvements (informed by XP-3A results)

In v2 (post-beachhead validation + compliance unlock conditions met)

- WhatsApp notifications (unlock: APP 8 legal review complete, Artifact 17 §07)
- Carer App in-app push notifications (unlock: APP 8 confirmation + Carer App build — largest COGS reduction lever, Artifact 20 §13)
- AlayaCare bi-directional integration (unlock: AX-02 confirmed, VI-1 validated)
- LLM-generated external content — personalised notifications (unlock: NFR-INJ-01–04 implemented + red-team tested)
- P-2 in matching algorithm (unlock: E-1 legal opinion obtained)
- Agency owner dashboard (coordinator performance, SPP completeness across agency)
- Family Contact portal (unlock: CC beachhead validated + E-3 gate live)

How long will it take?

This PRD does not set exact dates. The following phases give relative timeframes. All phases are gated by the conditions listed — not by calendar pressure.

Phase	What Happens	Condition to Start	Relative Length
Pre-Sprint Gate	AX-01 (CRIT-04 engineering confirmation) + Google Maps APP 8 review (SC-07) initiated	Engineering lead available	1 week
Sprint 1	Build Features 1–7. HIGH-06, HIGH-07, HIGH-08 resolved. Audit log 100% complete.	AX-01 confirmed, design wireframes approved by PM Lead	6–8 weeks

Phase	What Happens	Condition to Start	Relative Length
E1/E2 Concierge	PM Lead manually runs the product for Angela and Tom. OKR-1 through OKR-5 measured.	Sprint 1 PASS (harness-audit-grader verdict)	4 weeks
First Paid Trial	Agency Owner LOI converts to 30-day paid trial ($99 reduced entry).	OKR-1 AND OKR-3 validated at E1 (< 5 min fill time AND ≥ 40% 1-tap approval without calls)	30 days
v1 GA	First commercial subscription. SPP onboarding session priced.	OKR-1 through OKR-5 all validated. HIGH-01 through HIGH-05 resolved. DPIA-01–07 complete.	After trial close
v1.1	Compliance gap dashboard. SPP population improvements.	OKR-1 and OKR-3 validated post-trial.	~60 days post-GA

What comes before Sprint 1 starts?

The following are **pre-Sprint 1 gates** — the product cannot be coded until these are confirmed:

- **AX-01:** Engineer confirms in writing that `free_text_notes` field does not exist in the v1 SPP schema and no unstructured text field of any name exists in the v1 data model. (CRIT-04 completion.)
- **AX-02 result known:** Either bi-directional AlayaCare integration is scoped (v2 roadmap item) or scoping is deferred. Either answer is fine — the unknown is the problem.
- **Privacy Counsel briefed:** On DPIA-01 scope and timeline. Sprint 1 can begin before DPIAs are complete, but the counsel relationship must be active.
- **Design wireframes for 3-Tap Approval Flow (sketch-gate):** The Moment of Truth screen cannot be built from spec alone. Minimum acceptable fidelity: PM Lead-approved sketch showing element order, DR-1 font-size annotations, tap target dimensions ≥ 44×44px, and notification preview above Approve button. Full Figma polish required before E1 but not before Wave 4 coding start (Artifact 24 US-10 AC-6).
- **Copy Approval (B1-B):** Privacy Counsel confirms all four SMS template strings (ACT-C-01, ACT-C-02, ACT-P-01, ACT-F-01 — Artifact 23 §4.1) are APP-compliant. Angela (CC-001) has reviewed all four for coordinator and client clarity. Templates are committed to `config/sms_templates.py` with `approved_by` and `approved_date` fields set. No notification story passes `harness-audit-grader` SD-01B until this gate is closed.

9. Open Questions

ID	Question	Owner	Due	Impact if Wrong
AX-01	Does a free-text field of any name exist in the current v1 SPP schema?	Engineer	Before Sprint 1	CRITICAL — CRIT-04 compliance blocker
AX-02	Does AlayaCare Connect support write-back endpoints for assignment confirmation?	Engineer	2026-04-01	AlayaCare moat score, positioning claim, v2 integration scoping
SC-07	Does Google Maps Distance Matrix API DPA satisfy APP 8 — is only carer postcode transmitted (no client data)?	Legal	2026-04-01	ACT-V-03 cannot ship to production without this confirmed
V-1	Will Angela approve the top-ranked candidate without calling to verify — at first use?	E1 observation	E1 Week 1	If no: the product is a search tool, not a decision tool. Product-market fit is not confirmed.
VI-1	Will agency owners pay $150–250/month for a standalone product without EMR integration?	XP-2A	Q2 2026	If no: the entire go-to-market motion must shift. EMR integration required before launch — 6–9 month delay.
HIGH-01	What is the consent framework for collecting P-2, P-3, P-4, P-5?	Privacy Counsel	Before GA	Cannot ship without consent mechanism
HIGH-02	What is the lawful basis for disclosing carer SPP briefing (ACT-C-02)?	Privacy Counsel	Before GA	ACT-C-02 cannot ship without documented lawful basis
HIGH-03	DPIA for CC-1 compound (P-3 + P-4 + P-5) — when initiated?	Privacy Counsel	Sprint 1 initiation	Cannot collect compound CC-1 data without DPIA

10. Handshake Table — PRD → Execution Skills

ID	From PRD	To Next Skill	What Transfers
PRD-01	§7.2 Feature list + P1/P1.1/P2 classification	`prioritization-frameworks` (Execution Skill 2)	Feature list for opportunity scoring (Importance × 1-Satisfaction). Only ranked P1 features enter `agentic-logic-spec`.
PRD-02	§7.2 All P1 features + compliance constraints + CC guards	`agentic-logic-spec` (Execution Skill 3)	Features → pseudocode. Guards → IF-THEN gates. Constraints → NFRs. Threshold constants (E1_LEGAL_SIGNOFF, etc.)
PRD-03	§7.3 Model selection (Haiku/Sonnet/No LLM per action)	`agentic-logic-spec` §6 (Autonomy Classification)	Binding model assignment per action type. Any deviation = cost violation or CRIT-02 compliance violation.
PRD-04	§4 OKR-1 through OKR-6	`user-stories` acceptance criteria	Every P1 story's "Given/When/Then" must trace to a named OKR. A story with no OKR link is a Requirement Drift defect.
PRD-05	§7.4 Compliance constraints + CC-1 through CC-5 compound guards	`agentic-logic-spec` NFR section	Each compound combination becomes an explicit IF-THEN guard with a named `guard_id` in the pseudocode.

PRD integrity note: This document is the Source of Truth. Every downstream artifact — `agentic-logic-spec`, `user-stories`, `synthetic-phi-generator`, `harness-audit-grader` — traces back to a section of this PRD. If a downstream artifact contains a feature, constraint, or OKR not present here, it is a Requirement Drift defect. If this PRD changes, Stage A restarts.

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